

The Obstruction Spectrum of Weyl Context Families and the 2-adic Tower

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Abstract

State-independent contextuality of qudit stabilizer measurements is not just present or absent — it comes in a definite strength. For m -qudit Weyl–Heisenberg context families in local dimension d , the achievable all-versus-nothing obstruction value is exactly $\{0, d/2\}$ for every even d and $\{0\}$ for odd d . **This value classification — the even/odd dichotomy and the fixed value $d/2$ — is due to Abramsky, Cerelescu and Constantin [14]**, whose parity obstruction $2k \equiv 0 \pmod{d}$ (their Thm. 16) and normal-form classification (their Thm. 21) settle it; our Obstruction Spectrum Theorem (Theorem 1) recovers their $2S \equiv 0 \pmod{d}$ in explicit Weyl-certificate language. On this basis we develop three further layers. (i) A closed-form calculus: contextuality is decided by an integer commutator carry $b(C) = \sum_{\{u,v\}} \langle u, v \rangle_{\mathbb{Z}}/d \pmod{2}$ (Theorem 4) — no phases or matrices — with an exact value-bit formula (Theorem 6) and a cycle criterion (Theorem 8), in which the carry invariant Q has the algebraic form of a Pontryagin-square quadratic refinement (we do not construct the cohomology operation here). (ii) A 2-adic tower structure: the obstruction lives at the top layer and no single lift carries it (Depth Rigidity, Theorem 5), with a fiber doubling law governing its propagation $d \rightarrow 2d$ (generativity obstruction $G = \lfloor \Sigma M/2 \rfloor \oplus \lfloor K/2 \rfloor$). (iii) A three-device superconducting replication of the $d = 2$ witness. The obstruction’s geometric home — a canonical degree-2 class of order n on the stack $[X/\mathrm{PU}(n)]$ of maximal abelian subalgebras — is developed in the companion paper.

1 Setting and the question

Fix m qudits of local dimension d and symmetric Weyl lifts $W(v) = \tau^{-q(v)} X^{a_1} Z^{b_1} \otimes \dots$, $\tau = e^{i\pi/d}$, integer representatives $0..d-1$, $q(v) = \sum_i a_i b_i$. A *context* C is a word of pairwise-commuting v ’s ($\langle u, v \rangle \equiv 0 \pmod{d}$) with $\sum v \equiv 0 \pmod{d}$; its product is a scalar $\omega^{s(C)}$, $\omega = e^{2\pi i/d}$. A *certificate* is $\lambda \in \mathbb{Z}_d^{\mathrm{ctx}}$ with $\lambda^\top A \equiv 0 \pmod{d}$, where A is the context-by-observable multiplicity matrix; its value is $S = \lambda^\top s \pmod{d}$. A nonzero value is a state-independent all-versus-nothing (AvN) obstruction: no $\mathbb{Z}_d$ -valued noncontextual assignment is consistent with the family. Which values S can occur was settled by [14]; our aim is to recast that answer in explicit certificate language and add the carry calculus, the 2-adic tower, and a hardware replication. Let $H(d, m) \subseteq \mathbb{Z}_d$ be the set of achievable values.

Practical takeaway. Three facts are directly usable. (i) *Even/odd dichotomy*: state-independent Weyl contextuality exists exactly for even d ; in odd d every such family is noncontextual ($H = \{0\}$),

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so an experiment seeking a preparation-untrusted contextuality witness must use even local dimension. (ii) *Fixed strength*: when present, the obstruction takes the single value $d/2$ (the sign $-1 = \omega^{d/2}$) — there is no continuum of strengths to tune, and correspondingly no additive resource monotone (the obstruction is a saturating support, not a charge). (iii) *Cheap witnesses*: contextuality is decided by an integer commutator carry $b(C) = \sum_{i < j} \langle c_i, c_j \rangle / d \bmod 2$ with no matrices or phases, and a witness at $2d$ can be generated from one at d by the fiber construction (though genuine AvN minimality is a $\mathbb{Z}_d$ -cycle condition, Proposition 9, not the $\mathbb{Z}_2$ shadow). The natural application is state-independent certification of quantumness in even- d hardware: the witnesses hold for any input state, including the maximally mixed one, so they benchmark a device without trusting state preparation. We claim no bridge to code performance or T -count; the obstruction value classifies a cohomological invariant, not a computational resource.

Operational certification (application). The dichotomy gives a concrete measurement protocol. For a Weyl family one measures, in each context C , the joint outcomes of its commuting generators and forms the product statistic; the noncontextual functional $S = \sum_C \cos(2\pi(\hat{s}_C - s(C))/d)$ has a classical bound below the quantum value exactly when $H \neq \{0\}$, i.e. for even d . Two features make this device-friendly. First, the context product operator is the scalar $\omega^{s(C)}I$, so the statistic is fixed by the operator algebra and is independent of the input state (the maximally mixed state included); state-preparation error does not degrade it, and only measurement imperfection enters. A protocol caveat is essential here and is easily overlooked: the scalar identity $\prod_{O \in C} O = \omega^{s(C)}I$ means that a single joint measurement of a context—diagonalizing the commuting family and reading all outcomes from one basis label—returns the product $\omega^{s(C)}$ on every shot as an algebraic identity, independently of the device, so it certifies nothing (it reproduces the ideal value even on random outcomes). The measurement-limited character is recovered only when the observables of a context are measured in physically separate operations (sequential non-destructive readout, or one ancilla per observable): the outcomes are then no longer bound by the identity, and measurement error genuinely degrades the correlator below $|\omega^{s(C)}| = 1$. This preparation-untrusted, measurement-limited character of state-independent contextuality is well established experimentally on trapped ions, photons, and superconducting qubits [7, 8, 9]; we use it rather than claim it. Second, and specific to the present classification, the quantity being certified is not merely the presence of contextuality but its *value*: the graded invariant $S = d/2$, together with the even/odd signature (the same construction is provably noncontextual in odd d), is a structural fingerprint of the obstruction spectrum. A ququart-encoded ($d = 4$) witness realizing this on qubit hardware needs only shallow circuits and tolerates measurement error at the several-percent scale, well within current devices; the value-graded hardware certification is developed separately.

2 The Obstruction Spectrum Theorem

Theorem 1 (Obstruction Spectrum). *For every even d , every m , every Weyl context family and every certificate, $2S \equiv 0 \pmod{d}$; hence $S \in \{0, d/2\}$. For odd d (in the Gross convention $W'(v) = \omega^{-2^{-1}q(v)}T(v)$), $S = 0$ always. Thus the only state-independent obstruction value achievable by Weyl context families in any even dimension is $-1 = \omega^{d/2}$.*

Proof. The Weyl product obeys $W(u)W(v) = \tau^{c(u,v)}W((u+v) \bmod d)$ with $c(u, v) = -q(u) - q(v) + 2\beta(u, v) + q((u+v) \bmod d)$, $\beta(u, v) = \sum_i u_{b,i}v_{a,i}$ (machine-verified against matrices at $d = 4, 6, 9, 16, 25/25$ contexts each). Telescoping a context word from the identity, the q -terms of the partial sums

cancel:

$$2s(C) \equiv - \sum_i q(u_i) + 2 \sum_{j < i} \beta(u_j, u_i) \pmod{2d}, \quad (1)$$

$B(C) = \sum_i \beta(P_i, u_i)$, P_i the prefix sums; reducing mod d , bilinearity gives $B(C) \equiv \sum_{j < i} \beta(u_j, u_i)$. On commuting pairs $\beta(u, v) - \beta(v, u) = \langle u, v \rangle \equiv 0$, so β is symmetric mod d there; polarizing $\beta(\sigma, \sigma)$ at $\sigma = \sum_i u_i \equiv 0$, using $\beta(u, u) = q(u)$, yields $2 \sum_{j < i} \beta(u_j, u_i) \equiv - \sum_i q(u_i) \pmod{d}$. Substituting into (1) reduced mod d :

$$2s(C) \equiv -2 \sum_{u \in C} q(u) \pmod{d} :$$

the doubled phase is a coboundary with explicit potential $-q$. Therefore $2S = \lambda \cdot 2s \equiv -2(\lambda^\top A) \cdot q \equiv 0 \pmod{d}$. For odd d , 2 is invertible mod d and the Gross-convention telescoping gives $s(C) \equiv - \sum_{u \in C} q(u) \pmod{d}$ outright, so every certificate value vanishes. $\square$

Verification. The potential identity holds 0/3000 at each of $d = 2, 4, 6, 8, 12, 16$ and 0/18521 across mixed (d, m) ; the mod- $2d$ strengthening fails $\sim 50\%$ — the residue is exactly the obstruction bit, so the theorem is sharp. Odd- d triviality: single value $\{0\}$ at $d = 3, 5, 9$ (`spectrum_test2.py`; full ledger in the repository `INDEX.md`).

Corollary 2 (The obstruction is one bit). *$s \bmod (d/2)$ is exact with potential $-q$; all obstruction content lives in $b(C) := (s(C) + \sum_{u \in C} q(u)) / (d/2) \in \mathbb{Z}_2$, and $S = (d/2)(\bar{\lambda} \cdot b \bmod 2)$ with $\bar{\lambda} = \lambda \bmod 2$. Only $\lambda \bmod 2$ carries value; any attaining certificate has odd entries. The value group is always $\subseteq \mu_2 \subset \mu_d$: the physics factors through the real shadow at every even d , tying the two-tier structure of Paper A to a quantitative statement here.*

Remark 3 (No higher-order qudit parity proofs). Theorem 1 fixes the value in $\{0, d/2\}$ for any incidence structure, of any multiplicity: if each observable lies in r contexts, the concatenated word has every letter r times, yet the achievable value is still only 0 or $d/2$, independent of r . Since $d/2$ is the order-two element of μ_d (Corollary 2), there is no obstruction of order > 2 in any Weyl family — the value is always 2-primary and visible to the mod-2 shadow. In particular the natural expectation that high-multiplicity covers $(d \mid r, r > 2)$ would yield genuinely order- d “qudit parity proofs” fails: any such cover, if contextual, still carries the single bit $d/2$. (Verified: stacking a $d = 4$ certificate to multiplicities $r = 4, 6$ keeps the value group $\subseteq \{0, 2\}$, never $\{1, 3\}$.)

3 The commutator-carry form

Theorem 4 (Carry form). *Exactly, $2s(C) \equiv - \sum_i q(u_i) + 2 \sum_{j < i} \beta(u_j, u_i) \pmod{2d}$, and consequently*

$$b(C) \equiv \sum_{\{u, v\} \subseteq C} \langle u, v \rangle_{\mathbb{Z}/d} \pmod{2}$$

— the obstruction bit is the total commutator carry. The whole theory is combinatorial: no phases, no matrices, only symplectic carries.

Verification. Carry form 0/2500 at each $d = 2, 4, 6, 8, 12, 16$, and 0/800 per d with zero letters and high multiplicity.

4 Depth Rigidity

A lift of a level- d family to level $2d$ chooses $x_v \in \{0, 1\}^{2m}$ per observable ($\tilde{v} = v + d x_v$) keeping all contexts commuting and sum-zero mod $2d$ (a GF(2)-linear system with carry right-hand sides).

Theorem 5 (Depth Rigidity). *(a) If the commutation block of the lift system is solvable then $b(C) \equiv \sum_{v \in C} \langle x_v, v \rangle \pmod{2}$ is a coboundary, so every certificate value of the lifted family is 0. (b) Conversely, if a level- d family attains (some λ with value $d/2$), the $\bar{\lambda}$ -weighted commutation equations have identically zero left side and right side $\sum_j \bar{\lambda}_j b(C_j) \equiv 1$; hence attaining families never lift. The obstruction sits at the top 2-adic layer and is carried by no single section.*

Verification. On the $d = 8$ pool the full mod-8 kernel has value group $\{0, 4\}$ but the 2-divisible / liftable part gives only $\{0\}$; the $d = 8$ certificate λ is odd on 62/89 contexts. The dual no-lift witness for the $d = 8 \rightarrow 16$ step is machine-verified (data in the repository).

5 The value-bit formula

Theorem 6 (Value-bit formula). *For any mod- $2d$ kernel element $\tilde{\lambda}$ of a pure fiber pool, with odd support supp and lift data $u = \tilde{v} \bmod d$, $x = \tilde{v} \text{div } d$, the value bit P ($S = dP$) is*

$$P \equiv \frac{1}{2} \left[\sum_{\text{supp}} K + \sum_{\text{supp}} \Lambda(x) + (d/2) \sum_{\text{supp}} Q(x) \right] \pmod{2},$$

$K = \sum_{i < i'} \langle u_i, u_{i'} \rangle_{\mathbb{Z}/d}$, $\Lambda(x) = \langle \sum_i x_i, \sigma \rangle - \sum_i \langle x_i, u_i \rangle - 2 \sum_i \langle x_i, P_i \rangle$, $Q(x) = \sum_{i < i'} \langle x_i, x_{i'} \rangle$; the bracket is always even. The $(d/2)$ -terms are visible mod 2 only for $d \equiv 2 \pmod{4}$, reproducing the regime split.

Verification. 0/550 on each of $\text{PM} \rightarrow 4$, $\text{cert4} \rightarrow 8$, and non-attaining pools, non-vacuously (`Wformula.py`).

6 Attainment and the classification

The upper bound (Theorem 1) is complemented by attainment at every even d .

Theorem 7 (Attainment / classification). *For every even d and $m \geq 2$, the value $d/2$ is attained, so $H(d, m) = \{0, d/2\}$; for odd d , $H(d, m) = \{0\}$.*

For general even d the lower bound is the ROZF construction [2], whose even- d attainment we translate to Weyl certificates; the *generality* of attainment is inherited from that construction, while the explicit, self-contained certificates below — independently matrix-verified at $d = 8$ (value 4, 89 contexts) and $d = 16$ (value 8, 363 contexts) — are new. The ROZF face construction attains $\{0, d/2\}$ at $d = 4, 6, 8, 10, 12, 16, 20$; `verify_cert8.py` confirms the $d = 8$, $S = 4$ certificate on raw 64×64 matrices (commuting, scalar, $\lambda^T A \equiv 0 \pmod{8}$, value 4, system provably infeasible), and `verify_cert16.py` confirms $d = 16$, $S = 8$ on raw 256×256 matrices. The classification is thus a theorem, not contingent on the tower results of the next section; the $d = 16$ certificate in particular shows there is no depth ceiling.

7 A closed-form contextuality criterion

The value-bit machinery yields, for even d , an exact and elementary criterion for contextuality in terms of a single symplectic invariant of each cycle. Fix a family of contexts $\{C_r\}$, each a set of Weyl observables (symplectic vectors in $\mathbb{Z}_d^{2m}$) summing to 0 mod d ; let A be its $\mathbb{Z}_2$ context-observable incidence and $b_r = \sum_{i < j \in C_r} \langle c_i, c_j \rangle / d \pmod{2}$ the carry vector. A *cycle* is a left-kernel vector λ ($\lambda^\top A \equiv 0$), i.e. an even-multiplicity combination of contexts.

Theorem 8 (Cycle value = symplectic self-pairing). *Let d be even and λ a cycle. Then*

$$\lambda \cdot b = Q(\lambda) := \sum_{a < b \text{ in the observable multiset of } \lambda} \langle v_a, v_b \rangle / d \pmod{2}.$$

Consequently the family is contextual if and only if some cycle λ has $Q(\lambda)$ odd.

Proof. By definition $\lambda \cdot b = \sum_{r: \lambda_r=1} \sum_{i < j \in C_r} \langle c_i, c_j \rangle / d$, the intra-context pairs. Splitting Q into intra- and inter-context pairs, $Q = \text{intra} + \text{inter}$ with $\text{inter} = \sum_{a < b} \sum_{u \in C_a, v \in C_b} \langle u, v \rangle / d$. By bilinearity $\sum_{u \in C_a, v \in C_b} \langle u, v \rangle = \langle S_a, S_b \rangle$ where $S_r \in \mathbb{Z}^{2m}$ is the integer sum of the observables of C_r . Each context sums to 0 mod d , so $S_r = d t_r$ for an integer vector t_r ; hence $\langle S_a, S_b \rangle = d^2 \langle t_a, t_b \rangle$ and $\text{inter} = \sum_{a < b} d \langle t_a, t_b \rangle \equiv 0 \pmod{2}$ since d is even. Thus $Q = \text{intra} = \lambda \cdot b$. The family is contextual iff $b \notin \text{rowspan}(A)$, i.e. iff some left-kernel λ has $\lambda \cdot b = 1$, i.e. $Q(\lambda)$ odd. $\square$

The invariant Q has the algebraic form of a Pontryagin-square quadratic refinement of the cycle: contextuality is the oddness of this quantity on some cycle. This is the sense in which the value obstruction resembles the support of a type-III (Pontryagin) anomaly of the Weyl–Heisenberg group — the obstruction saturates rather than adds under stacking. We do not here construct the cochain complex, coefficient sequence, and cohomology operation that would identify Q with the Pontryagin square on the nose; that precise identification is left open (see §10). Generic cycles have $Q = 0$ (verified 300/300 on random families); attaining certificates are precisely those carrying a cycle with $Q = 1$. For odd d the halving step fails and the family is noncontextual ($H = \{0\}$), consistent with the absence of odd- Q cycles.

Proposition 9 (The exact $\mathbb{Z}_d$ criterion; Q is its shadow). *Let M be the $\mathbb{Z}_d$ incidence matrix and $\gamma(C) = \sum_{i < j} \langle c_i, c_j \rangle / d \in \mathbb{Z}_d$ the context scalar. By the Fredholm alternative over $\mathbb{Z}_d$, a family is state-independent contextual ($\gamma \notin \text{col}(M)$) if and only if there is a $\mathbb{Z}_d$ -cycle λ ($\lambda^\top M \equiv 0 \pmod{d}$) with $\lambda \cdot \gamma \not\equiv 0 \pmod{d}$. The odd- Q criterion of Theorem 8 is the mod-2 shadow of this. For even d the shadow is sound, and the proof is two lines: if $Mx \equiv \gamma \pmod{d}$ then also $Mx \equiv \gamma \pmod{2}$, so any λ with $\lambda^\top M \equiv 0 \pmod{2}$ has $\lambda \cdot \gamma \equiv \lambda^\top Mx \equiv 0 \pmod{2}$. Hence the shadow can never fire on a noncontextual system: odd- Q has no false positives at any even d .*

It is, however, incomplete, and in the opposite direction: a $\mathbb{Z}_d$ -cycle certifies contextuality when $\lambda \cdot \gamma \not\equiv 0 \pmod{d}$, and if that value is nonzero but even its mod-2 reduction vanishes, so the shadow stays silent on a genuinely contextual system. An explicit non-degenerate $d = 8$ witness is pinned in `shadow_soundness_exact.py`.

Remark 10 (Correction to v1). Version 1 of this paper stated the gap backwards. It claimed that a minimal odd- Q support “can carry the $\mathbb{Z}_2$ shadow obstruction without closing over $\mathbb{Z}_d$ ”, cited “an explicit false certificate at $d = 8 \dots$ recorded in the repository”, and reported that 28.6% of random $d = 8$ incidence systems carrying a $\mathbb{Z}_2$ -odd cycle are not genuinely $\mathbb{Z}_d$ -contextual. All three are wrong. The false-positive rate is not 28.6% but exactly 0, by the proof above; no such certificate exists to be recorded, which is why the cited artifact was never committed; and the true defect is

the missed-detection direction described above. The error made the criterion look weaker than it is. What survives is the sentence that followed it, on certificate minimization: a minimal odd- Q support need not be a $\mathbb{Z}_d$ -cycle, so it can be a false *witness* even though the *verdict* is always right. Version 1 conflated witness with verdict.

Why Q is nonetheless a good criterion: a Weyl-specific fact. On generic $\mathbb{Z}_d$ scenarios the mod-2 shadow is a poor proxy — not because it lies, which it cannot, but because it is often blind: under the protocol of `shadow_soundness_exact.py` it misses roughly 39% of genuinely $\mathbb{Z}_8$ -contextual random incidence systems. On Weyl families, by contrast, the $\mathbb{Z}_2$ shadow and the true $\mathbb{Z}_d$ obstruction agree generically (300/300 on random families); the discrepancy appears only on engineered minimal cycle supports. Thus the completeness of the odd- Q criterion is not linear algebra but a consequence of the symplectic structure of the Weyl group, which forces the carry shadow to be faithful. For certificate minimization one must still search over $\mathbb{Z}_d$ -cycles, since minimality is where the shadow can fail.

8 The 2-adic tower (structural)

Beyond the classification lies the question of how the obstruction is generated across dimensions. The explicit certificates were all found by a uniform construction — the pure fiber pool over the minimal attaining family one level below (all local lifts of every base context) — and the obstruction propagates up the 2-adic tower $\text{PM} \rightarrow d=4 \rightarrow d=8 \rightarrow d=16$ through fibers, never through sections (Theorem 5). We record what is proved and what remains.

Theorem 11 (Fiber projection). *For $\tilde{\lambda}$ in the mod- $2d$ kernel of a fiber pool $\text{Fib}(F)$, the projection $\pi(\tilde{\lambda})_j = \sum_x \tilde{\lambda}_{j,x}$ satisfies $\pi^\top A_{\text{base}} \equiv 0 \pmod{2d}$ and is everywhere even; hence $\mu = \pi/2$ is a mod- d base kernel element. Consequently the value group of $\text{Fib}(F)$ at $2d$ is $\subseteq 2 \cdot (\text{value group of } F)$ under $\mathbb{Z}_d \hookrightarrow \mathbb{Z}_{2d}$.*

Conjecture 12 (Fiber doubling law, $m_0 \geq 4$). *For a base family F at even level $m_0 \geq 4$ outside a generative subclass, the value-group generators obey $g_{\text{fiber}} = 2g_{\text{base}}$. The reflect direction (F attains $\Rightarrow \text{Fib}(F)$ attains) holds on every attaining family tested; the preserve direction (F trivial $\Rightarrow \text{Fib}(F)$ trivial) fails on a positive fraction of trivial bases even at $m_0 = 4$ (see below), so the law holds exactly on the non-generative subclass, characterized in closed form below.*

Status. Verified 51/51 across attaining anchors (PM, cert4, ROZF(d), $d \leq 10$) and random families, and 10/10 on ROZF families $d = 4..22$. The reflect direction is solid on every attaining family tested. Preserve, however, leaks: pre-filtering random bases by actual base-triviality, a positive fraction (empirically $\sim 12\text{--}15\%$ at $m_0 = 4, 6$) of trivial bases have an attaining fiber. So the remaining obligation is not a blanket δ -vanishing lemma but a characterization of the non-generative trivial bases on which every fiber certificate has value 0. Via Theorems 6, 11 this is a finite GF(2) statement, and this session localizes the obstruction precisely to the carry of $\frac{1}{2}(T_A + T_B)$ in the value-bit formula (the base-commutator piece T_A and the high quadratic $(d/2)T_Q$ both vanish for $4 \mid m_0$; integer halving does not distribute over $T_A + T_B$, and that carry is the entire content). [open]

The generativity mechanism, made exact. Using the criterion of Theorem 8, the fiber cycle value decomposes exactly. Writing each fiber observable as $\tilde{v} = u + dx$ ($u \in \mathbb{Z}_d$ the base residue, $x \in \{0, 1\}$ the lift bit), the fiber Q is

$$Q_{\text{fiber}} = Q_{\text{base-residue}} \oplus \sum_{a < b} \left[\frac{\langle u_a, u_b \rangle + d M_{ab} + d^2 \langle x_a, x_b \rangle}{2d} \right] - \left\lfloor \frac{\langle u_a, u_b \rangle}{2d} \right\rfloor \pmod{2},$$

with $M_{ab} = \langle u_a, x_b \rangle + \langle x_a, u_b \rangle$ (the underbraced term is the lift-induced halving carry). Generativity is precisely the event that this lift-induced carry is odd while the base is trivial. Crucially it is a floor-carry, not a quadratic form in the lift bits — which is why an Arf/quadratic treatment of the doubling law does not close it: the obstruction is of the same carry type as the value bit b itself. This carry now has a closed form on all fiber cycles. Writing $\Sigma M = \sum_{a < b} M_{ab}$ and $K = \#\{a < b : M_{ab} \text{ odd}\}$,

$$G = \lfloor \frac{\Sigma M}{2} \rfloor \oplus \lfloor \frac{K}{2} \rfloor \pmod{2},$$

verified exactly (7500/7500 at $d = 4$, 1200/1200 at $d = 6$). The first term is the intrinsic pairing $\Sigma M = \langle U, X \rangle - \sum_i \langle u_i, x_i \rangle$ of the cycle's total base-residue $U = \sum u_i$ and total lift-bit $X = \sum x_i$; the second is the carry produced by summing the odd cross-terms (the sum-of-floors versus floor-of-sum defect), which vanishes exactly when K is even — in particular on repeat-free cycles, recovering the intrinsic criterion $\langle U, X \rangle - \sum_i \langle u_i, x_i \rangle \equiv 0, 1 \pmod{4}$. The generativity/doubling-law obstruction is thus fully characterized: a cycle is non-generative iff $G = 0$.

The carry decomposition is provable as an exact integer identity. For any integers $M_1, \dots, M_N$ with $S = \sum_a M_a$ and $K = \#\{a : M_a \text{ odd}\}$,

$$\sum_a \lfloor \frac{M_a}{2} \rfloor = \lfloor \frac{S}{2} \rfloor \oplus \lfloor \frac{K}{2} \rfloor \pmod{2},$$

since $\lfloor M/2 \rfloor = (M - (M \bmod 2))/2$ gives $\sum_a \lfloor M_a/2 \rfloor = (S - K)/2$, while $S \equiv K \pmod{2}$ gives $\lfloor S/2 \rfloor = (S - (K \bmod 2))/2$, and the difference is $-\lfloor K/2 \rfloor$. Hence the closed form for G is proved once Q_{fiber} is the sum of the per-pair halving carries $\lfloor M_{ab}/2 \rfloor$ (the one remaining verified geometric step); the arithmetic that turns the per-pair carries into the two-term formula is exact.

Remark 13 (Generativity and contextuality activation). The fiber can be generative: a base with trivial value group can have an attaining fiber, creating contextuality absent below. This is strongest at $m_0 = 2$ but, correcting an earlier expectation, persists at $m_0 = 4, 6$ at a $\sim 12\text{--}15\%$ rate per random trivial base; generativity is a generic tower phenomenon, not a qubit-only artifact. Read as a protocol, this is contextuality activation by dimension embedding: a passive embedding $\mathbb{Z}_d \hookrightarrow \mathbb{Z}_{2d}$ (a free operation — no entanglement or magic) converts a noncontextual family into a state-independent contextual one. The yield is tunable by the number of base contexts k , rising monotonically $5.6\% (k=3) \rightarrow 76.4\% (k=12)$ (250 trials per point, Wilson 95% CI), plausibly $\rightarrow 1$ for large k . Because the witness is state-independent, the certified nonclassicality holds for any input state, including the maximally mixed one — a preparation-assumption-free feature. Crucially, generativity is a single-context fiber effect and is structurally distinct from the multi-context certificate cycles that carry the value- $d/2$ obstruction (see Section 10).

Connection to quasiprobability negativity. The equivalence between (measurement) non-contextuality and the existence of a nonnegative quasiprobability representation is a general theorem [6, 10]; on the quasiprobability side, no nonnegative diagram-preserving representation exists in any even dimension [10], which is the even/odd dichotomy seen here from the dual direction. Our contribution in this language is constructive and quantitative: the odd- $Q / \mathbb{Z}_d$ -cycle criterion is a closed-form, decidable test for whether a Weyl measurement family admits such a representation (the general equivalence is non-constructive), and the value classification refines the binary present/absent statement to a fixed obstruction strength $d/2$. We add no new simulation bound: the state-independent obstruction places even- d Weyl measurement families on the negative-representation side of the measurement axis, but classical simulation cost is governed by state negativity, a distinct quantity.

9 Related work

Existence of parity/AvN obstructions and the cohomological criterion “proof exists $\iff [\beta] \neq 0$ ” are due to Okay–Roberts–Bartlett–Raussendorf [1]; our certificate framework repackages the existence side. Even- d cohomological obstructions and general- d attainment are ROZF [2]; odd- d noncontextuality and the $2^{-1} \bmod d$ mechanism are Gross and Raussendorf et al. [3, 4]. Linear-algebra enumeration of $\mathbb{Z}_2$ parity proofs (product $\pm I$ only, no $\mathbb{Z}_d$ spectra, no towers) is Lisoněk–Raussendorf–Singh [5]. The formulation of observable-based contextuality as the unsolvability of a GF(2) linear system, with the qubit Pauli group encoded in the symplectic polar space W_n and a numerically computed degree of contextuality (minimum unsatisfied weight), is the Saniga–Holweck–de Boutray–Giorgetti program [11, 12]; it is qubit-only ($d = 2$) and computes the decision by SAT/heuristic search. Our even- d commutator-carry b and incidence matrix A are the $\mathbb{Z}_d$ lift of their W_n encoding; the distinction is that our obstruction admits a closed-form per-cycle evaluation (Theorem 8) rather than a min-weight search, and extends to even qudits with a value spectrum. Conversely, their quantitative degree of contextuality $d_H(E, \text{Im } A)$ lifts verbatim to even d through our carry vector b : it counts how many context constraints must be flipped to restore a valuation, and is a distinct, orthogonal axis to the value classification (a family of degree 1 can carry the maximal value $d/2$). A qudit contextuality-degree over $\mathbb{Z}_d$ (rather than the $\mathbb{Z}_2$ coset weight) is a natural open refinement. A complete characterization of which qubit stabilizer states admit an AvN proof (the AvN-triple theorem: every such argument reduces to a GHZ-equivalent triple) is Abramsky–Barbosa–Carù–Perdrix [17], with completeness / false-negative analysis by Carù and Abramsky et al. [18]; a state-independent AvN class via partial closure of commuting observables is Kim–Abramsky [13]. **The even/odd dichotomy and the value classification $\{0, d/2\}$ are due to Abramsky–Cercelescu–Constantin [14]** (parity obstruction $2k \equiv 0$, their Thm. 16; commutation-group normal-form classification, their Thm. 21), who also give a $\mathbb{Z}_2 \rightarrow \mathbb{Z}_{2k}$ transfer construction. Our Theorem 1 recovers their result in Weyl-certificate language; the contribution here is the explicit carry/value-bit calculus, the 2-adic tower (Depth Rigidity and the doubling law), and the hardware, with the geometric/cohomological account of the obstruction in the companion paper. All of this AvN literature works over $\mathbb{Z}_2$ (qubit stabilizers, product $\pm I$) and characterizes AvN existence. Our distinct contributions are of a different kind: the even- d value classification $H(d, m) = \{0, d/2\}$ (a spectrum over $\mathbb{Z}_d$, not existence over $\mathbb{Z}_2$); the closed commutator-carry form (Theorem 4); Depth Rigidity with a certificate-derived no-lift witness (Theorem 5); the explicit high- d certificates; and the fiber-tower program.

Relation to Abramsky–Cercelescu–Constantin: dictionary and logical status. Under the identifications

<i>this paper (Weyl realization)</i>	<i>Abramsky–Cercelescu–Constantin [14]</i>
Weyl label v	commutation-group generator / vector
symplectic form $\langle u, v \rangle$	commutator matrix entry $\mu(x, y) = v_{x,y}$
closed context C	commuting subword / relation
certificate λ	contextual word
value S	central phase k
carry $b(C)$	commutation factor $\sum \mathcal{I}(w)$ (inversion sum)
$\mathbb{Z}_d$ value assignment	left splitting / noncontextual homomorphism

Theorem 1 is the Weyl–Heisenberg *specialization* of their Theorem 16: their inversion sum $\sum \mathcal{I}(w)$ is our carry accumulation, their polarization $2 \sum \mathcal{I}(w) = \sum_{x < y} n_x n_y v_{yx}$ is our polarization of β , and their $2k \equiv 0 \pmod{d}$ is our $2S \equiv 0 \pmod{d}$. Our spectrum theorem is therefore neither independent of nor a strengthening of theirs; it re-derives their classification in the Weyl realization,

and the carry $b(C)$ is their commutation factor in symplectic-form coordinates — a change of realization and an explicit closed form, not a new invariant. What is *not* present in [14] and is offered here as the candidate new content: the explicit value-bit closed form for 2-adically lifted certificates (Theorem 6); the *cross-dimensional* tower statements — Depth Rigidity (Theorem 5) and the doubling law — which concern relations *between* d and $2d$ rather than the parity constraint at a single d (and must be read as such, not as restatements of it); the hardware replication; and the geometric/cohomological home (companion paper). A full theorem-by-theorem equivalence with [14]’s normal-form invariants is the natural next step. The extended arXiv version of [14] additionally provides a Darboux-normal-form reduction and a closed-form relative-parity criterion for the existence of state-independent contextuality in general commutation groups; that existence classification is complementary to the value spectrum and state-sector facet geometry developed here. Information-theoretic “cost of contextuality” measures [15, 16] quantify simulation overhead rather than the cohomological obstruction value studied here.

10 Open problems

(1) The δ -vanishing lemma, closing Conjecture 12 for $m_0 \geq 4$. The closed-form criterion (Theorem 8) settles the static contextuality question exactly — contextual iff some cycle has odd Q — and clarifies the tower question: propagation $d \rightarrow 2d$ is now the statement that the odd- Q property transfers from base cycles to fiber cycles. The certificate quadratic form q is genuine (third differences vanish, verified) and on a multi-cycle space the fiber-attainment map is quadratic with nonzero polarization, so an Arf invariant governs the fiber transfer. We caution that an earlier “multi-cycle $\Rightarrow$ contextual” expectation is false: by Theorem 8, generic cycles have $Q = 0$ and are noncontextual (verified 300/300 on random $m=2$ families); contextuality requires the special odd- Q condition. (2) The activation subclass of Remark 13 as a protocol (the generativity carry G is now closed; what remains is its operational reach). (3) Ramification necessity: do attaining families always live on the mod- d colliding locus? (both known minimal attaining families do). (4) Minimize the $d = 16$ certificate. (5) AvNR/Carù-style completeness of the admissible-cycle certificates.

The geometric step of the doubling law. We record the identity $Q_{\text{fiber}} \equiv \sum_{a < b} \lfloor M_{ab}/2 \rfloor \pmod{2}$ together with its proof, as it completes the value-transfer arithmetic. Writing each level- $2d$ observable as $v = u + dx$ (u the base representative mod d , $x \in \{0,1\}^{2m}$ the lift bits) and $q_u = \lfloor \langle u_a, u_b \rangle / d \rfloor$, $M_{ab} = \langle u_a, x_b \rangle + \langle x_a, u_b \rangle$, $s_{xx} = \langle x_a, x_b \rangle$, the exact per-pair carry is

$$\left\lfloor \frac{\langle v_a, v_b \rangle}{2d} \right\rfloor \equiv \left\lfloor \frac{q_u + M_{ab} + d s_{xx}}{2} \right\rfloor \pmod{2}$$

(a pure floor identity, since $0 \leq \langle u_a, u_b \rangle \pmod{d} < d$ prevents the remainder from carrying; verified exactly at $d = 4, 6, 8$). For even d this splits as $\lfloor q_u/2 \rfloor + \lfloor M_{ab}/2 \rfloor + \frac{d}{2}s_{xx} + (q_u \pmod{2})(M_{ab} \pmod{2})$, so

$$Q_{\text{fiber}} - \sum_{a < b} \lfloor \frac{M_{ab}}{2} \rfloor \equiv \sum_{a < b} \lfloor \frac{q_u}{2} \rfloor + \frac{d}{2}s_{xx} + (q_u \pmod{2})(M_{ab} \pmod{2}) \pmod{2}.$$

The first term $\sum \lfloor q_u/2 \rfloor$ depends only on the base u ’s and is identically zero as a base-context vector (checked for the minimal attaining families at $d = 4, 6, 8$), so it vanishes on every fiber. The term $\frac{d}{2} \sum_{a < b} s_{xx}$ closes in general even d : it is nonzero only for $d \equiv 2 \pmod{4}$, and there the polarization identity for the quadratic refinement $Q(x) = \sum_i x_{2i} x_{2i+1}$ gives $\sum_{a < b} \langle x_a, x_b \rangle = Q(r_C) + \sum_{v \in C} Q(x_v)$ (using the fiber closure $\sum_{v \in C} x_v = r_C$, $r_C = \frac{1}{d} \sum_{v \in C} u_v \pmod{2}$), so the term

is the explicit per-observable coboundary $\phi(v) = \frac{d}{2}Q(x_v)$ plus the base-context constant $\frac{d}{2}Q(r_C)$, itself a base coboundary (verified $d = 4, 6, 10, 14$). The last term $\sum_{a < b} (q_u \bmod 2)(M_{ab} \bmod 2)$ reorganizes as $\sum_v \langle w_v, x_v \rangle$ with $w_c = \sum_{a \neq c} (q_{u_a} \bmod 2) u_a$: a per-observable coboundary of linear form in the lift bits, whose $\mathbb{F}_2$ -system is solvable with no base-context correction ($t \equiv 0$), verified at $d = 4, 6, 8, 10, 12, 14$; hence this term too lies in the column space of A and pairs to zero with every cycle. This solvability holds for all even d , which completes the geometric step. Split by $d \bmod 4$: for $d \equiv 0 \pmod{4}$ the base symbol $u = \frac{d}{2}p$ has $\frac{d}{2}$ even, hence $u \equiv 0 \pmod{2}$ componentwise; since M_{ab} is bilinear, $M_{ab} \equiv 0 \pmod{2}$ on every fiber context, so the term $\equiv 0$ identically. For $d \equiv 2 \pmod{4}$, $\frac{d}{2}$ is odd, so $u \equiv p \pmod{2}$ and $M_{ab} \bmod 2$ depends only on the fixed base symbols p and the lift bits x , not on d ; labeling observables and contexts by their d -independent (symbol, lift) signature, the coboundary matrix A and the target vector T_{mix} are identical for every $d \equiv 2 \pmod{4}$ — one fixed 768×144 $\mathbb{F}_2$ system (byte-identical for $d = 6, 10, 14, 18, 22, 26, 30$, weight $|T_{\text{mix}}| = 128$). Solvability is therefore a single fact about this fixed system, established at $d = 6$, and transfers verbatim to every $d \equiv 2 \pmod{4}$. The complementary class $d \equiv 0 \pmod{4}$ is settled separately, and more easily: writing $u = hp \bmod d$ with $h = d/2$, bilinearity gives $q_u = \text{symp}(u_a, u_b)/d = hs$ where $\text{symp}(p_a, p_b) = 2s$, so h even forces every q_u to be even and T_{mix} to vanish *identically*, pointwise, with no cocycle summation required. The whole system is thus a function of $d \bmod 4$ alone, constant within each class and vanishing outright on one of them (verified $d = 4, \dots, 20$; see `mod4_dichotomy.py`). Hence T_{mix} is a coboundary for all even d ; together with the vanishing of $\sum \lfloor q_u/2 \rfloor$ and the explicit potential $\phi(v) = \frac{d}{2}Q(x_v)$ for the s_{xx} term, the geometric step of the doubling law is established for all even d .

11 Hardware validation at the base of the tower ($d = 2$)

The $d = 2$ endpoint of the spectrum—value $d/2 = 1$, i.e. the Peres–Mermin obstruction—was executed on a superconducting device (IBM `ibm_marrakesh`) to validate the protocol of the preceding remark. Each of the six contexts was realized as a three-observable sequential non-destructive measurement (one shared ancilla, Hadamard-test readout of each commuting Pauli into its own bit, with reset between), so that the per-context correlator is not algebraically pinned and genuinely responds to device error. The Cabello witness $S = \langle R_1 \rangle + \langle R_2 \rangle + \langle R_3 \rangle + \langle C_1 \rangle + \langle C_2 \rangle - \langle C_3 \rangle$ has noncontextual bound 4 and quantum value 6. On each device we pinned all six contexts to a single low-error qubit triple (selected by readout- and two-qubit-gate error), enabled dynamical decoupling and measurement twirling, and took 8192 shots per context. Replicating the identical protocol across three IBM Heron devices gave, per device,

$$S_{\text{marrakesh}} = 4.94 \pm 0.02, \quad S_{\text{fez}} = 4.56 \pm 0.02, \quad S_{\text{kingston}} = 4.74 \pm 0.02,$$

each individually violating the bound by $32\text{--}61\sigma$, with an inverse-variance-weighted combination

$$S = 4.761 \pm 0.010 \quad (80\sigma, \text{ shot-noise only}),$$

and per-context magnitudes $0.74\text{--}0.85$ (e.g. $\langle C_3 \rangle$ ranging -0.74 to -0.82), consistent with reported superconducting realizations. This combined figure is shot-noise-only: the three devices are *not* mutually consistent (a homogeneity test gives $\chi^2 \approx 266$ on 2 degrees of freedom), so once the chip-to-chip scatter is honored a systematics-aware (Birge-scaled) combination gives $\approx 7\sigma$. The robust statement is therefore the *per-device* violation ($32\text{--}61\sigma$, shot noise); the 80σ combined number should not be read as a single- S significance, and no nondisturbance/compatibility diagnostic has been performed (repository `hardware/COMPATIBILITY.md`). The 0.38 spread in S across devices is genuine chip-to-chip variation and leaves every device far above the classical bound. An initial run

in which contexts were transpiled independently placed one context on poorer qubits and yielded a marginal $S = 4.06 \pm 0.03$ (2.0σ); the fixed-layout, error-suppressed, multi-device figure above is the reliable one. We report this only as a validation of the measurement methodology at the base of the tower; it certifies the presence of the $d = 2$ obstruction, not the value grading, whose hardware realization at $d = 4$ we leave to the separate value-graded study. We note that, by Corollary 2, the obstruction phase equals $e^{2\pi i(d/2)/d} = -1$ at every even d , so higher- d hardware witnesses exhibit the same -1 signature: the d -dependence of the spectrum is an algebraic ($\mathbb{Z}_d$) datum, certified in software, rather than a distinct analog phase on-device.

A Computational certificates

All load-bearing certificates ship in the companion repository, with each claim mapped to a script and its pinned one-line output in INDEX.md: `spectrum_test2.py` (potential identity, odd- d Gross convention); `Wformula.py` (Theorem 6); `verify_cert8.py`, `verify_cert16.py` (raw-matrix $d = 8, 16$ certificates); `criterion.py` (the closed-form odd- Q criterion); `tmix_dindep.py`, `close_T2_proof.py` (the fixed $\mathbb{F}_2$ system and the geometric step of the doubling law); `kernelpk.py`, `howell.py` (Smith/Howell-form solvers, brute-force unit-tested). The additional stress tests reported inline (large-sample potential-identity and carry-form scans, the ROZF attainment sweep $d \leq 20$, the fiber doubling-law statistics, and the $d = 8$ false-certificate example) are archived in the same repository ledger.

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